Muchen Yan MA BEnvs

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Muchen Yan is an architect and PhD candidate at the University of Innsbruck, researching digital design and fabrication for circular construction. Her work combines computational design, robotic fabrication and AI-assisted workflows to turn reclaimed materials such as demolition concrete and timber into new architectural components, tested through research-practice projects in Innsbruck. Earlier research covered human-robot collaborative prototyping, augmented reality and bio-based materials.

EDUCATION

2024-present	**University of Innsbruck** | PhD-Doctoral programme Architecture
2020-2022	**Royal Danish Academy (KADK)** | Master of Computation in Architecture – Focus: human-robot collaboration, augmented reality, digital fabrication
2015-2018	**University of Melbourne** | Bachelor of Environments, Major in Architecture

ACADEMIC EXPERIENCE

04/2024-present — **University of Innsbruck** | PhD Candidate Dissertation Position
Innsbruck

- Conduct research on digital design and fabrication for circular construction, with a focus on concrete upcycling strategies and design methods
- Teach seminars and workshops on computational design and materialisation (see Teaching)
- Lead the research-practice project Arbor+: Robotic fabricated reciprocal structure, Innsbruck (2026)
- Led the research-practice project Reuse Garden: Upcycling Concrete-Timber Waste to Urban Furniture, Innsbruck (2024-2025)

02/2023-03/2023 — **CITA / Politecnico di Milano** | Course Assistant / Tutor
Part-time, Copenhagen / Milan

- Preparing and teaching *Experimental Design of Hybrid Textile Bending Active Systems*, part of the Integrated Course of Advanced Design Studio

2021-2022 — **CITA (Centre for Information Technology and Architecture)** | Research Assistant
Part-time, Copenhagen

- Co-developed and built a 15 × 20 m textile installation at ZKM | Center for Art and Media Karlsruhe, Germany, contributing to design development, model building, digital fabrication and on-site construction
- Operated and maintained workshop machinery (3D printers, laser cutter, CNC machine, wood and metal workshop)

TEACHING

Seminars and workshops, University of Innsbruck

WS 2026/27	**Specialization structure and design: Prototyping** | Master Course Reuse Intelligence: AI for circularity
SS 2026	**Structure and Ecology** | Master Course Robotic fabrication for reciprocal structures
SS 2024-2026	**Methods of Materialization** | Bachelor Course Computational design methods with reused materials
WS 2025/26	**Methods in Construction and Design** | Bachelor Course Digital design for bio-integrative structures and robotic prototyping
SS 2025	**Specialization structure and design: Prototyping** | Master Course Digital fabrication and construction
WS 2024/25	**Structure and Ecology** | Bachelor Course Computational design, robotic prototyping and bio-integration

GRANTS & AWARDS

2026	**Early Stage Funding** | Project leader and applicant *Waste Quarrying: Developing an Action-Oriented Framework for Cascading Reuse of Demolition Concrete*
2015-2018	**Melbourne Global Scholarship** | University of Melbourne

PUBLICATIONS

2026	**Yan, M.**,Schinegger, K., & Rutzinger, S. (2026). *Hybrid Reclaimed Systems: Upcycling Concrete-timber Structures for Multidimensional Design Challenges*. In: Proceedings of the IASS-IWSS Conference 2026. In press
2026	**Yan, M.**,Schinegger, K., & Rutzinger, S. (2026). *Rubbles: Material Features as Design Data*. In: Proceedings of the EKA Arh Symposium 2026. In press
2026	**Yan, M.**,Schinegger, K., & Rutzinger, S. (2026). *Construction Waste Rubble Reimagined to Urban Furniture – Design-to-fabrication*. Poster, Future of Construction 2026.
2021	**Yan, M.**, & Tamke, M. (2021). *Augmented Reality for Experience-centered Spatial Design: A Quantitative Assessment Method for Architectural Space*. In: Towards a New, Configurable Architecture – Proceedings of the 39th eCAADe Conference, Vol. 1, Novi Sad, Serbia, pp. 173-180

EXHIBITIONS

2022	**ADFÆRD / VELFÆRD** | Brain Scans in VR and Reality *Royal Danish Academy, Copenhagen, Denmark* – Produced 3D scans as student specialist
2021	**70% Less CO_2** | Predictive Response *Royal Danish Academy, Copenhagen, Denmark* – Developed robotic 3D printing of a hybrid bio-based material as student researcher
2021	**BIOMEDIA** *ZKM | Center for Art and Media Karlsruhe, Germany* – Developed robotic 3D printing of a hybrid bio-based material as student researcher
2017	**Intelligent Ornamentation** | AA Visiting School Shanghai (student participant) *China 3D Printing Culture Museum, Shanghai, China*
2016	**Bodega, Ecological Metabolism** | AA Visiting School at IAAC (student participant) *IAAC, Barcelona, Spain*

RELATED PROJECTS

2026	**Arbor+: Robotic fabricated reciprocal structure** *i.sd Structure and Design, Innsbruck*
2024-2025	**Reuse Garden: Upcycling Concrete-Timber Waste to Urban Furniture** *i.sd Structure and Design, Innsbruck*
2022	**Reviving Crafting in Digital Design with Human-Robotic Collaborative Prototyping** *CITA, Royal Danish Academy, Copenhagen, Denmark*
2021	**Augmented Reality for Experience-centered Spatial Design** *CITA, Royal Danish Academy, Copenhagen, Denmark*